

Biotransformation of Sugarcane Bagasse Extract into High-Value Dextran: A Microbial Valorization Approach

Internship Project Report

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BT22BTECH11012

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भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

May 2025

Abstract

The exopolysaccharide, dextran, was prepared by culturing *Leuconostoc mesenteroides* in sucrose rich media. The global demand for dextran is increasing due to their broad-spectrum applications in biomedical and hydrogel industries. To meet this demand and to not have a dilemma with food safety, agri-waste materials with high sucrose content can be used as feedstock materials. The current study explored the ability of sugarcane bagasse (SCB) to provide sucrose for dextran production using *L. mesenteroides*. The effect of temperature, feedstock concentration, and time on sucrose extraction from SCB was studied to yield maximum sucrose. The effect of different combinations of salts on the production of dextran from commercial grade sucrose as well as SCB sucrose was evaluated. The salts, MnSO_4 , H_2O , CaCl_2 , and NaCl were selected for the study and it was found that a combination of the selected three salts and a fermentation period of four days yielded maximum dextran; 0.2821 g/g commercial grade sucrose and 0.661 g/g SCB sucrose. Dextran presence in the fermentation precipitate was confirmed using FT-IR spectroscopy, while its crystalline structure was assessed through XRD analysis.

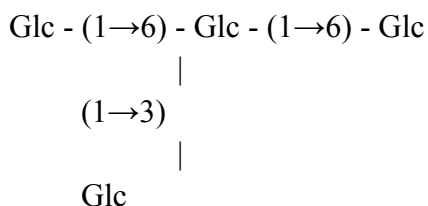
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1. Introduction

Dextran is a complex polysaccharide composed entirely of glucose units. It's main chain is made up of D-glucose repeating units connected by α -(1→6) glycosidic Linkages, where side chains are predominantly attached through α -(1→3) glycosidic bond. Sporadically, the side branches are linked through α -(1→2) and α -(1→4) glycosidic linkages depending on the bacterial strain and fermentation conditions (Sarwat et al., 2008). This branching pattern gives dextran its unique physicochemical properties such as high water solubility, varying viscosity, and potential for chemical modification due to the presence of multiple hydroxyl groups.



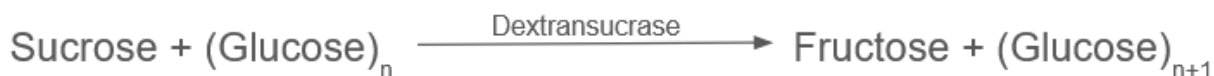
Dextran is produced by various lactic acid bacterial strains (LABs) such as *Lactobacillus*, *Leuconostoc*, *Weissella*, *Streptococcus*, *Pediococcus*, and *Acetobacter*. Among these bacteria, *L. mesenteroides* is the most effective strains used to produce dextran at the industry scale from sucrose as a sole carbon source due to the suitable structural configuration (95% α -(1→6) glycosidic bonds in backbone with 5% α -(1→3) glycosidic branched chains (Sarwat et al., 2008, Rahman et al., 2020).

Dextran's molecular mass, thermal stability and structural behaviour is quite important from the industrial application point of view. The Molecular mass of dextran can vary significantly from 40 kDa to 2000 kDa on a commercial level. For example, dextran-40 kDa and dextran-70 kDa are commonly used in medical applications such as blood plasma expanders due to their appropriate size and biocompatibility (Rahman et al., 2020). Due to its low antigenicity, biodegradability, high water solubility, and high thermal stability with a melting point of 278.36 °C (Rahman et al., 2020), it is widely used in other industrial applications like blood flow improver, petroleum recovery, adjuvants, emulsifiers, stabilizers, thickeners for jam and ice cream, improving moisture retention, food appearance and flavor, cross-linked dextran for protein purification and separation, magnetic resonance imaging enhancer, anticoagulant, preventing sugar crystallization, and as sweeteners (Rahman et al., 2020).

The biological synthesis of dextran is carried out by the enzyme called dextransucrase, which is secreted by *Leuconostoc* bacteria. This enzyme catalyzes the transfer of glucose from sucrose to a growing dextran chain. During the reaction, Sucrose is split, and the glucose is added to the polymer (Dextran), and Fructose is released as a by-product (Sarwat et al., 2008, Rahman et al., 2020). In addition to sucrose as carbon source, the fermentation media has to be provided with

N-source and certain trace elements essential for *L. mesenteroides* growth and subsequent dextransucrase activity. It was reported that CaCl_2 (Chae et al., 2009) and $\text{MnSO}_4 \cdot \text{H}_2\text{O}$ (Huang et al., 1976) acts as an activator in the production of dextransucrase whereas NaCl has shown to improve dextran yield.

Reaction is:



In the total cost of production, nearly 50-60% of the cost is used in the fermentation step of dextran production (Onilude et al., 2013). For that reason, Researchers uses various strategies: (i) selection of easily available and cost-effective feedstock; (ii) genetically engineered strains; (iii) optimization of medium formulation and culturing condition; (iv) mode of fermentation; and (vi) residue for the improvement of dextran production under economical aspect. Low-cost substrates that can be used for the production of microbial dextran are beet molasses (Miljković et al., 2016), tomato juice supplemented with sucrose (Han et al., 2014), wheat bran extract (Kajala et al., 2016), date seeds (Moosavi-Nasab et al., 2020) and rice bran extract (Shibayama et al., 2018). The present study is using Sugarcane Bagasse (SCB) as a substrate to produce dextran since it's cheap, and an abundant agro-industrial waste rich in sucrose.

Objectives:

1. Extraction of Sucrose from Sugarcane Bagasse (SCB)
2. Microbial dextran production using *Leuconostoc mesenteroides*
3. Dextran purification and characterization

2. Materials and Methods

2.1 Microorganisms and raw materials collected

The microbial strain used in this experiment *Leuconostoc mesenteroides* (NCIM 2073) was obtained from NCIM(National Collection of Industrial Management), pune, India. The strain was maintained in Glycerol stock (25%) at 4°C. Sugarcane Bagasse(SCB) was collected from the local market in Sangareddy, Telangana, India.

2.2 Microbial Kinetics study of *L. mesenteroides*

The growth kinetics of *L. mesenteroides* was studied by culturing them in MRS broth at 25°C and 100 rpm and the corresponding OD was measured at 600 nm with 3h intervals.

2.3 Extraction of Sucrose from Sugarcane Bagasse

Sugarcane bagasse was cut into short pieces, oven-dried, and ground to 0.25 mm sieve size. Sucrose was extracted from SCB powder using water as a solvent. The influence of process parameters: temperature, feedstock concentration, and extraction time was studied using One Variable At a Time (OVAT) approach. The concentration of sucrose was estimated using modified Anthrone method (Finley et al., 1973). The porosity % of SCB after extraction was calculated using the Tea-Bag Method and it was compared with SCB before extraction.

2.4 Synthesis, Recovery, Purification and Characterization of Dextran

The fermentation media was prepared using the following composition (g/mL): Yeast Extract, 0.01 and K_2HPO_4 , 0.03 and its pH should be in between 7 - 8 (Tripathy et al., 2024). The effect of commercial sucrose and SCB extract as a carbon source. The Minimal Media was made using 3 different salts: $CaCl_2$, NaCl, and $MnSO_4 \cdot H_2O$. The fermentation media was sterilized at 121°C for 15 min and cooled to 25°C. The fermentation media was inoculated by *L. mesenteroides* culture grown in MRS media till OD at 600 nm reaches 0.9 ($\sim 10^9$ CFU/ml). The Fermentation media was kept in an orbital shaker at 25°C and 100 rpm shaking speed. The influence of the salts; $CaCl_2$, NaCl, and $MnSO_4 \cdot H_2O$ and fermentation time was studied on dextran production.

After Fermentation, the broth containing dextran was centrifuged at 10000 rpm for 5 min to remove cell debris and the collected supernatant was concentrated in the oven. Dextran was precipitated by adding chilled ethanol, 2-fold the amount of concentrate and recovered by centrifuging at 10000 rpm at 5 min. The supernatant was discarded from the sample and the obtained precipitate was resuspended in distilled water, followed by precipitation with chilled ethanol (2-fold amount) to purify dextran. The purified dextran was recovered by centrifuging the samples at 10000 rpm for 5 min and oven-dried (Aman et al., 2012). The purified dextran was characterized using Fourier Transform Infrared (FT-IR) spectroscopy and X-ray Diffraction (XRD) analysis.

3. Results & Discussion

3.1. Microbial Kinetics study of *L. mesenteroides*

The growth kinetics of *L. mesenteroides* was studied by correlating the OD at 600nm to cell density in MRS media. The MRS broth was inoculated by secondary culture of *L. mesenteroides* with 1% inoculum volume. From Fig 1 it was observed that the log phase of *L. mesenteroides* lies between 12h to 24h. Hence, the microbial cell density at OD of 0.9 was selected for further studies.

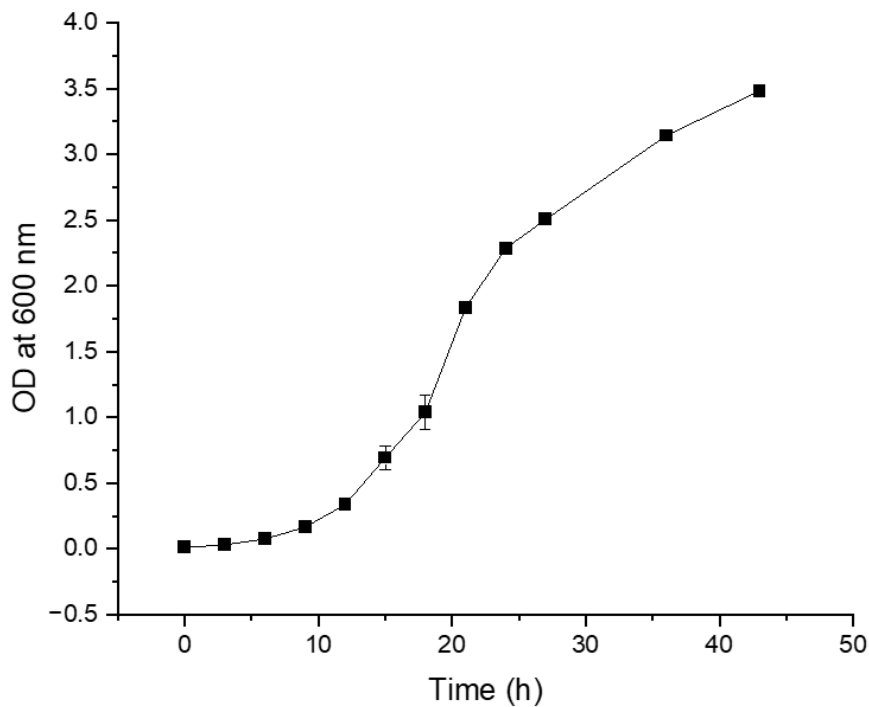


Fig 1 The growth kinetics of *L. mesenteroides* in MRS media

3.2. Extraction of Sucrose from Sugarcane Bagasse

3.2.1 Effect of Temperature

The experiment to evaluate the effect of temperature was conducted at 5%(w/v) of feedstock. It was observed that at high temperature, a higher amount of sucrose was extracted from SCB (Fig 2). This can be due to the increased solubility of sucrose and hydrolysis and possible degradation of structural integrity of SCB components. The deterioration of SCB structure causes it to be more porous facilitating the diffusion of sucrose to the liquid. Hence, sucrose extraction at 90°C was used for further experiments.

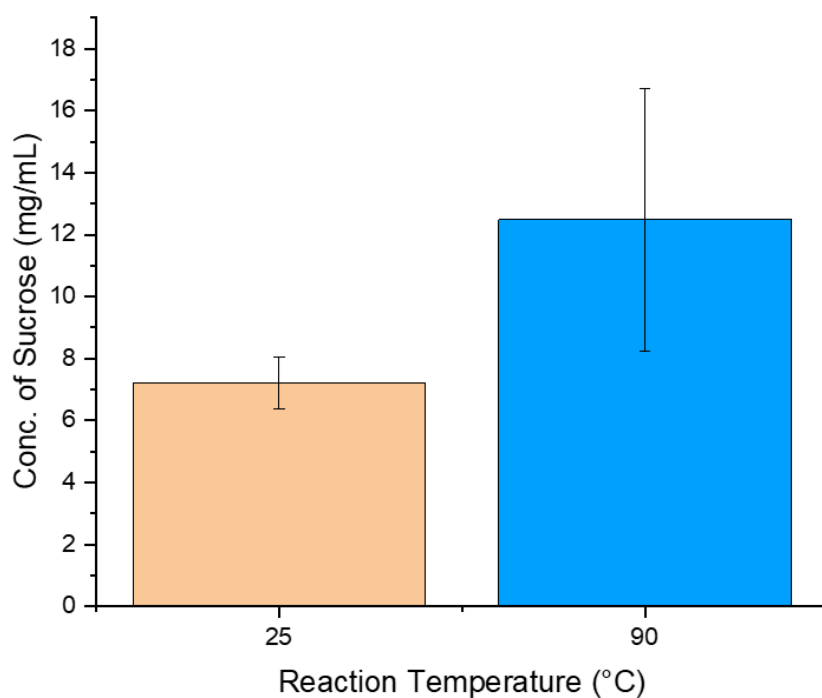


Fig 2 Effect of Temperature on extraction of sucrose from SCB

3.2.2 Effect of SCB Concentration

The experiment was carried out at 90°C for 1 h to evaluate the effect of SCB concentration to produce sucrose. The effect SCB concentration was varied from 2 - 10% of feedstock to evaluate the difference in sucrose concentration. It was observed that higher concentration of SCB resulted in higher sucrose concentration, but sucrose yield was getting saturated after 4% SCB concentration (Fig 3). This can be due to the entrapment of sucrose in the interior of SCB and the unavailability of enough solvent to facilitate sucrose diffusion to the liquid phase. Hence, for further studies 4% SCB concentration was selected.

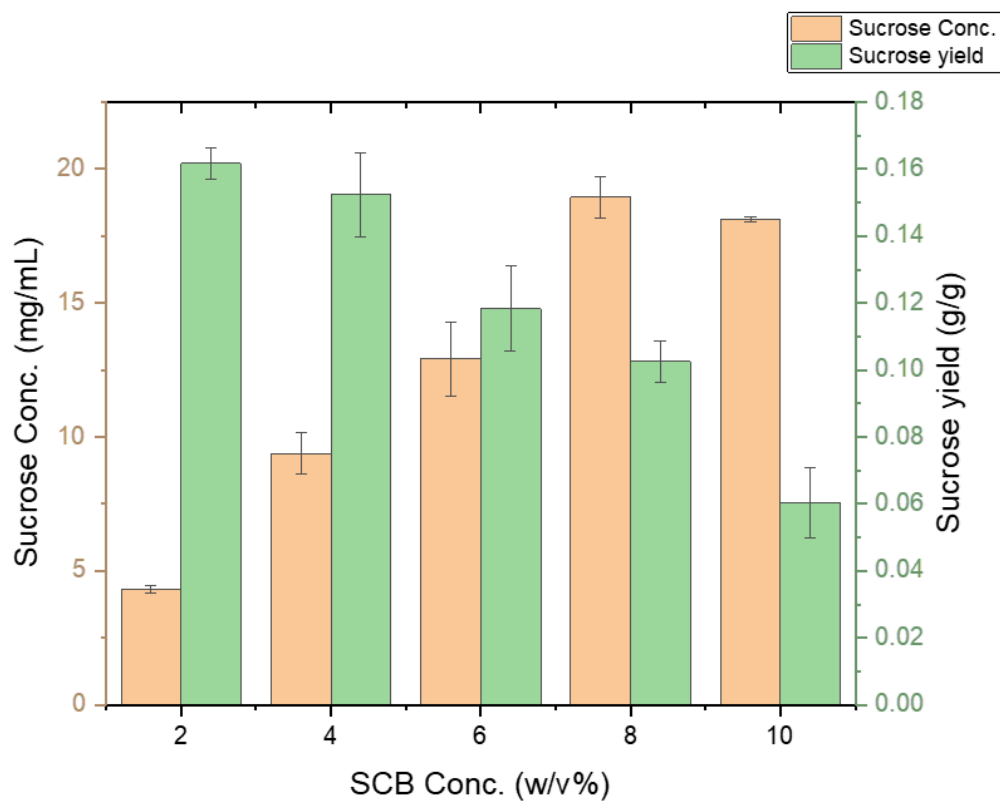


Fig 3 Effect of SCB concentration on extraction of sucrose from SCB

3.2.3 Effect of Reaction Time

The experiment was done to study the effect of time for the extraction of sucrose from 4% SCB concentration at 90°C over a range of 10-50 mins. It was observed that after 30 mins of incubation sucrose concentration has saturated, indicating a maximum removal of sucrose from SCB (Fig 4).

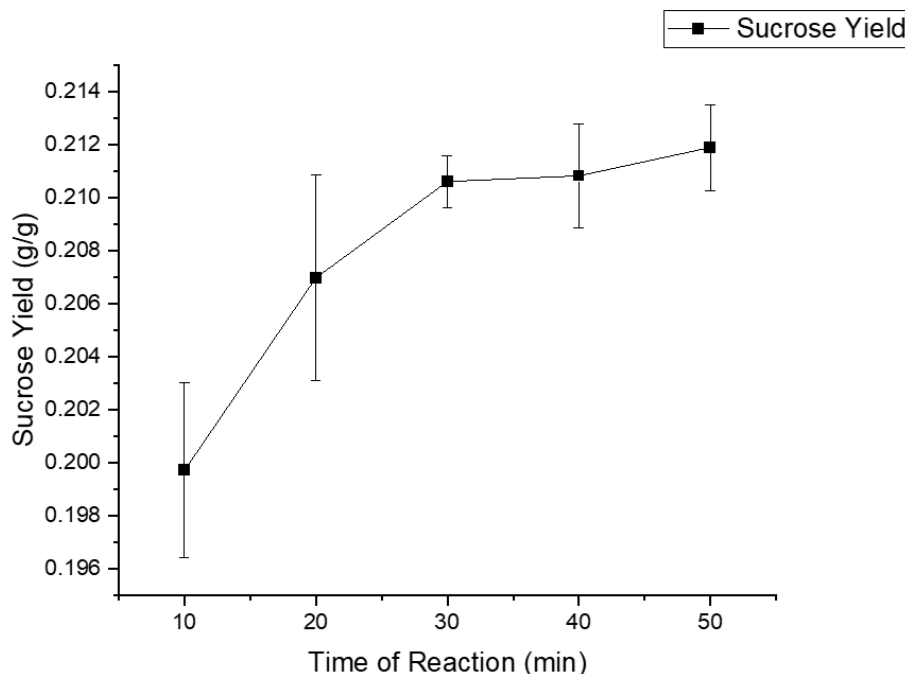


Fig 4 Effect of time on extraction of sucrose from SCB

Hence for further studies, aqueous extraction of sucrose was conducted at 90°C and 4% SCB concentration for 30 mins. The porosity % of SCB after sucrose extraction was found to be 25.78% whereas it was 18.67% for SCB before extraction. This shows that extraction of sucrose increased the porosity of feedstock by 38.08% and since sucrose and glucose is not a part of SCB's structural components, the elevation in porosity is acceptable.

The maximum yield of sucrose from SCB as per Wang et al., 2022 was found to be 0.33g/g of SCB, hence the current extraction technique has an efficiency of approx~ 85%.

3.3. Synthesis, Recovery and Purification of Dextran:

Dextran was produced using *L. mesenteroides* in commercial grade sucrose and SCB sucrose media (Fig 5). A visual examination showed that the dextran from commercial grade sucrose is whiter than the dextran from SCB sucrose, suggesting SCB dextran needs to be washed with ethanol further. The influence of presence of $\text{MnSO}_4 \cdot \text{H}_2\text{O}$, CaCl_2 , and NaCl and fermentation time on dextran yield were evaluated.



Fig 5 Commercial grade dextran in comparison with dextran from commercial grade sucrose and SCB sucrose

3.3.1 Effect of salts

The individual effects of salts on the production of dextran was studied at various concentrations and it was observed that $\text{MnSO}_4 \cdot \text{H}_2\text{O}$ is having the highest impact on dextran yield (Fig 6). A set of experiments were conducted on the effect of combination of salts on dextran yield (Fig 7) and it was observed that the two combinations $\text{NaCl} + \text{CaCl}_2$ and $\text{MnSO}_4 \cdot \text{H}_2\text{O} + \text{NaCl}$ yielded a comparative amount of dextran. The experiment with salt combination $\text{MnSO}_4 \cdot \text{H}_2\text{O}$ and CaCl_2 were unable to perform due the precipitation of the salts. However, a combination of the three showed a higher dextran yield than any of the individual or dual combinations of the salts. The result suggests the synergistic effect of the three salts: $\text{MnSO}_4 \cdot \text{H}_2\text{O}$, CaCl_2 , and NaCl are required to facilitate enzymatic production of dextran. Wang et al., 2024 observed that Mn^{2+} and Cl^- ions are essential for the activity of dextransucrase; they act as inducers for dextran production. Hence, they cannot be evaluated as single salts.

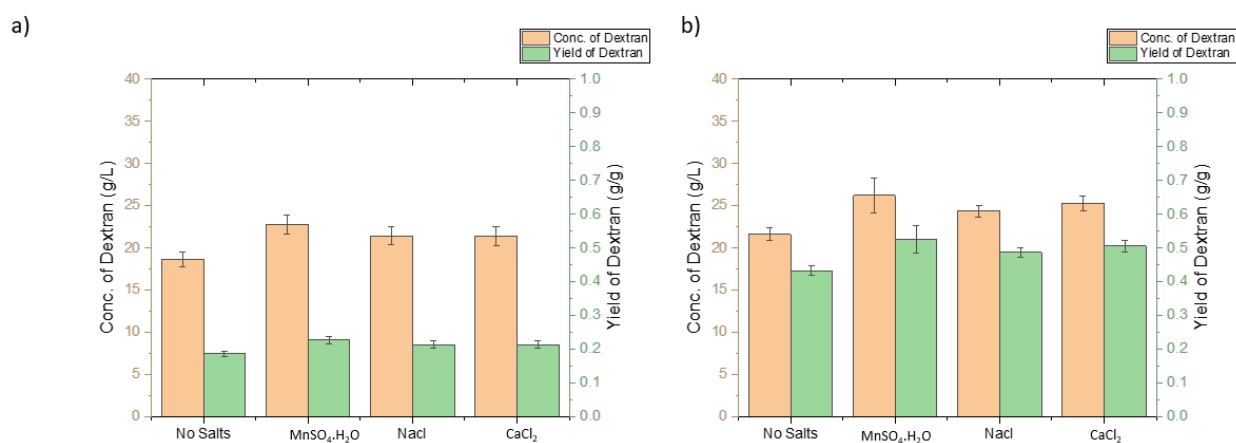


Fig 6 Effect of single salts on the production of dextran (a) from commercial grade sucrose (b) from SCB sucrose

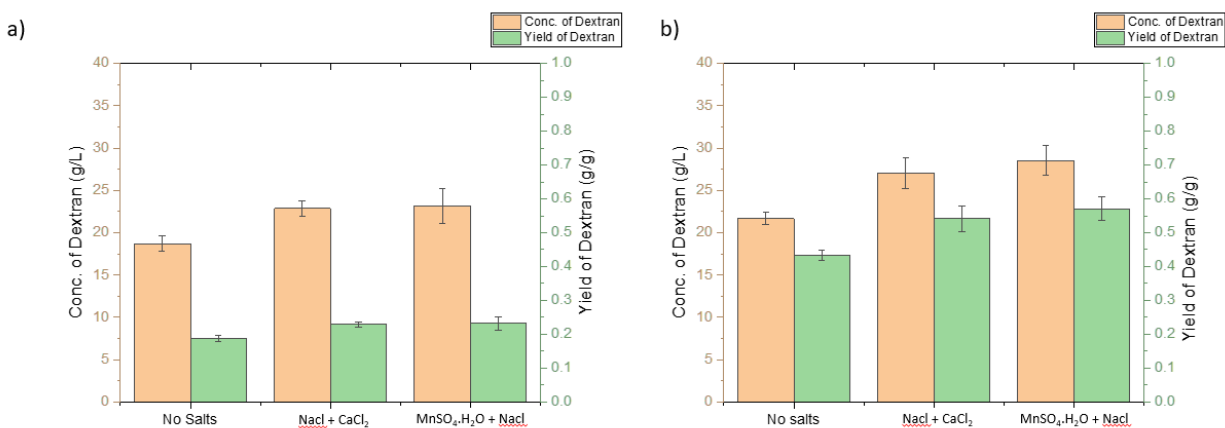


Fig 7 Effect of combination of salts on production of dextran (a) from commercial grade sucrose (b) from SCB sucrose

3.3.2 Effect of Fermentation time

The effect of fermentation time on the yield of dextran (Fig 8) showed that 4 days of fermentation yielded higher dextran in both cases: 0.2821 g/g commercial grade sucrose and 0.661 g/g SCB sucrose. However, further studies are required to find out the optimum fermentation time for maximum dextran production.

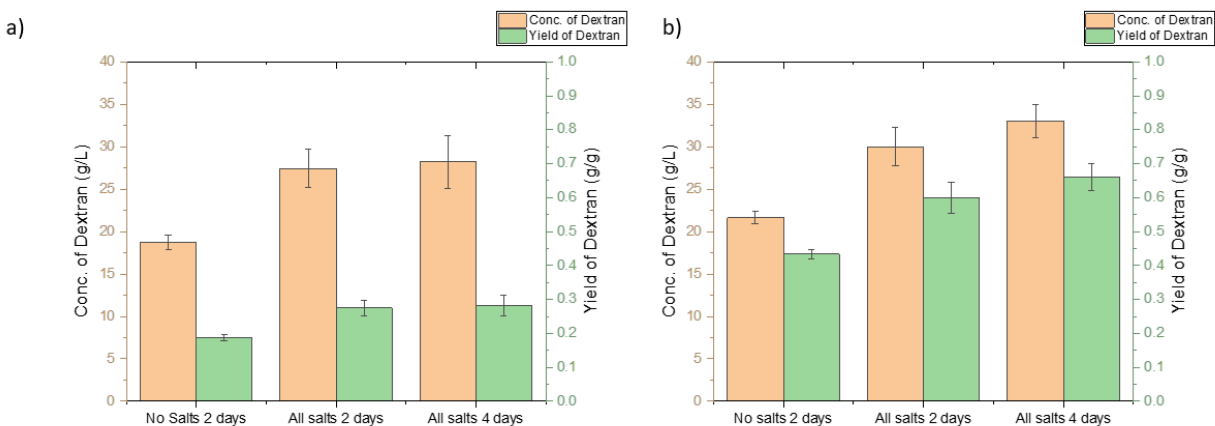
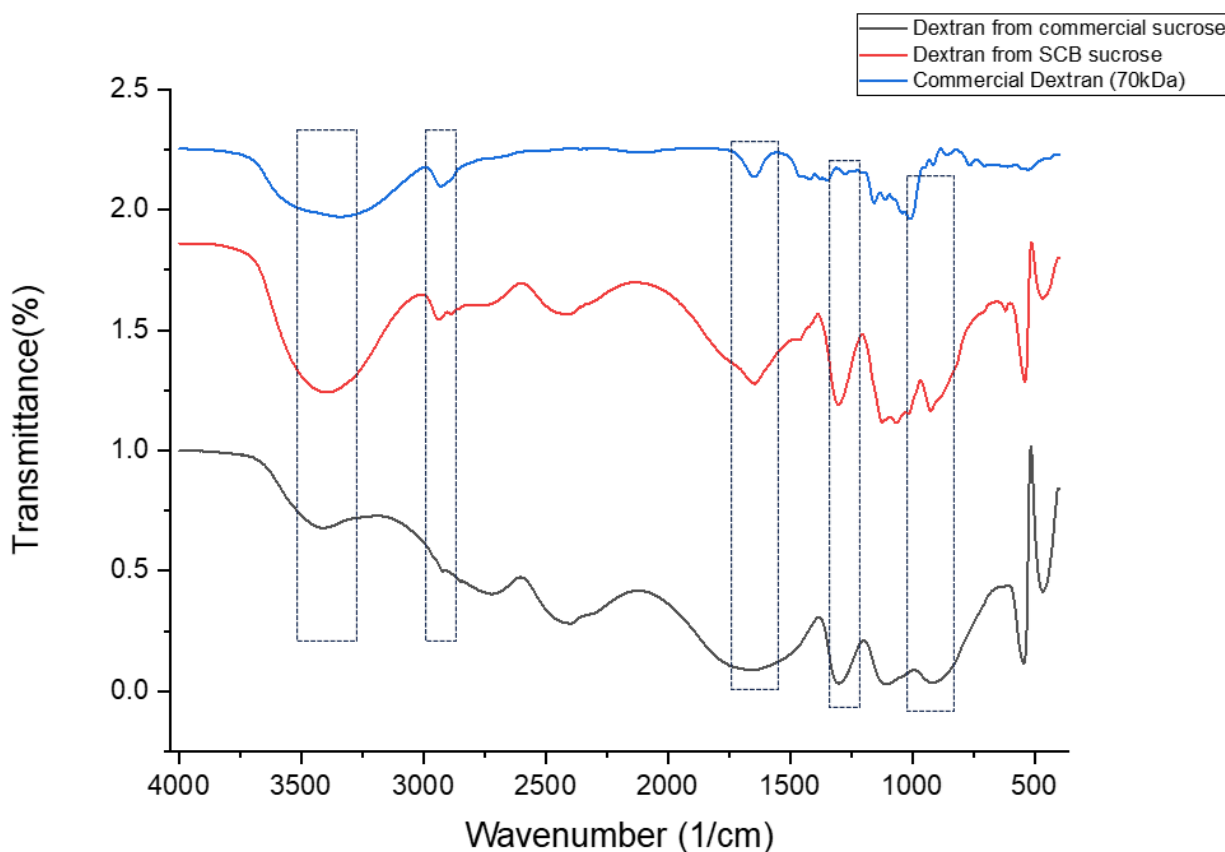


Fig 8 Effect of fermentation time on production of dextran (a) from commercial grade sucrose (b) from SCB sucrose

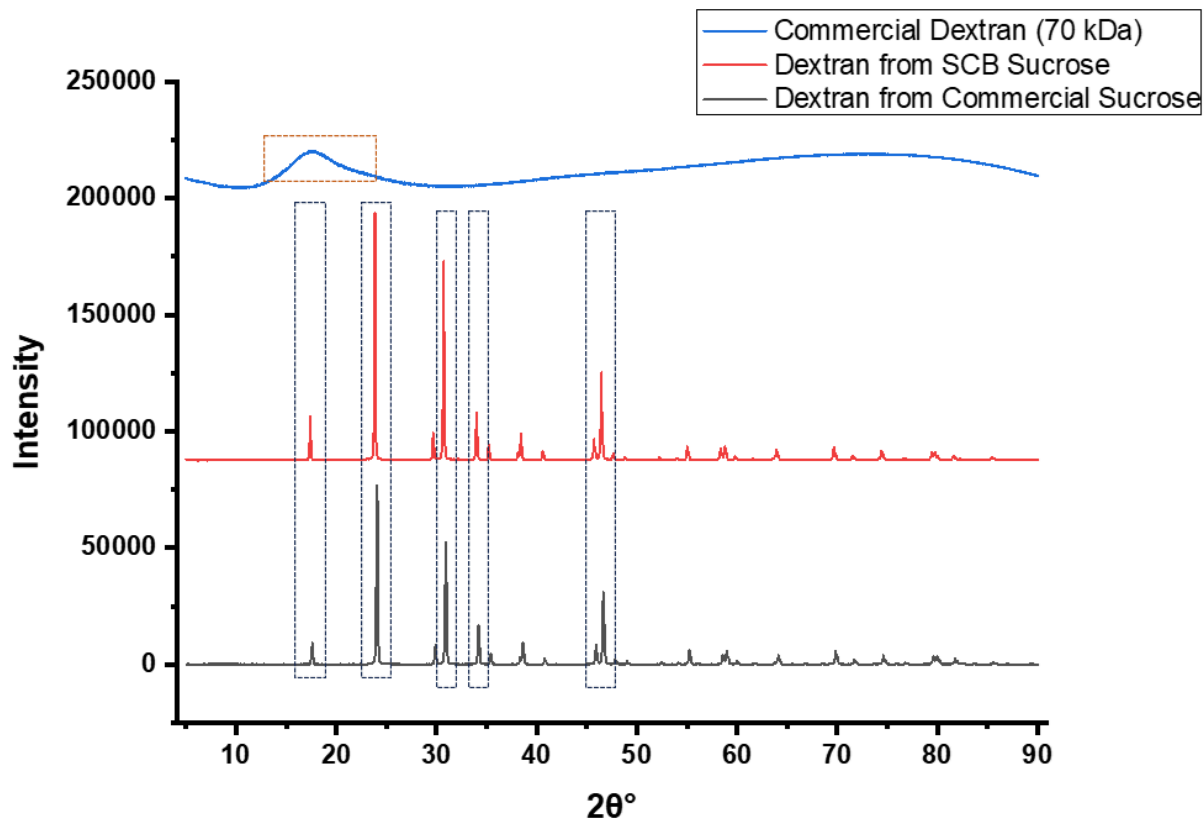
Characterization of Synthesized Dextran:

FT-IR analysis was done to identify the functional groups present in the dextran structure. The peak in between $3400\text{--}3500\text{ cm}^{-1}$ shows the presence of O-H groups (Kedaria et al ., 2017). The absorption peak at 2925 cm^{-1} and 1636 cm^{-1} were attributed to C-H stretchings (Paulo et al ., 2012) which suggests that the recovered compound was a polysaccharide. A distinct peak at 1153 cm^{-1} was characteristic of the glycosidic bond vibrations (C-O-C). The absorption peak at 1018 cm^{-1} was linked to the α -(1–6) glycosidic bonds in the dextran backbone (Zhang et al ., 2024). Hence, it can be confirmed that dextran was present in the precipitate obtained after fermentation.



XRD analysis was performed to determine the crystalline nature of the used compound. There were 5 sharp peaks at $2\theta \sim 17^\circ, 24^\circ, 31^\circ, 34^\circ$ and 47° observed in the XRD chromatogram suggesting crystalline structure of recovered dextran (both commercial sucrose dextran and SCB dextran) (Yilmaz et al ., 2022). Whereas for commercial dextran there is only one peak at $2\theta \sim 17^\circ$ that showed its amorphous nature (Du et al ., 2022). This highly crystalline structure can be a result of oven-drying compared to the commercial dextran which is spray dried. Crystallinity

properties can improve the mechanical strength, and thermal stability of the dextran. In addition to this, non-toxicity, bioactive property, and biodegradability can expand dextran's application in the biomedical industry as a hydrogels, drug carriers, in plasma volume expansion, and in implants (Jiang et al ., 2023).



4. Conclusions

In this study, the aqueous extract of SCB was used as a sucrose source for the production of dextran. It was observed that *L. mesenteroides* can survive and produce dextran in SCB aqueous extract. The comparison study showed that the yield of dextran from SCB sucrose was comparable to commercial grade sucrose. However, a visual analysis on the produced dextran shows that the dextran produced from SCB sucrose needs to be purified further. Dextran was identified in the precipitate after fermentation through FT-IR spectroscopy and the crystallinity was evaluated by XRD analysis. Hence, further studies on fermentation and purification are required for optimum dextran production from SCB sucrose.

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